

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2174

Acec Quincy 138 kV
July 30, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2174. The Interconnection Customer proposes a 154.5 MW solar photovoltaic facility interconnecting at the Acec Quincy 138 kV substation in the Virginia area. The request was received on August 31, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 741 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,811,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Acec Quincy – Ni Riv Dp 138.0 kV
Contingency: Parsons St – Acec Quincy 138.0 kV
Loading: 216.5% (Rating: 161 MVA)
DFAX: 100.0%
Pre-project loading: 120.2%
2. Facility: Acec Quincy Substation – Parsons Street 138.0 kV
Contingency: Acec Quincy – Ni Riv Dp 138.0 kV line
Loading: 173.8% (Rating: 200 MVA)
DFAX: 100.0%
Pre-project loading: 96.5%
3. Facility: Parsons St – Todds Tavern Dp 138.0 kV
Contingency: Acec Quincy – Ni Riv Dp 138.0 kV
Loading: 139.5% (Rating: 219 MVA)
DFAX: 100.0%
Pre-project loading: 68.8%
4. Facility: NiRiver Dp – Chancellor Substation 138.0 kV
Contingency: North Anna 345/138kV Transformer Bank
Loading: 137.7% (Rating: 224 MVA)
DFAX: 79.3%
Pre-project loading: 83.1%
5. Facility: Gulfgate – Story City Substation 138.0 kV
Contingency: N Anna 345 kV/138 kV Transformer
Loading: 112.5% (Rating: 211 MVA)
DFAX: 37.6%
Pre-project loading: 85.0%

6. Facility: Chancellor Substation – Story City 138.0 kV
 Contingency: N Anna 345-138 kV XFMR
 Loading: 107.8% (Rating: 421 MVA)
 DFAX: 79.3%
 Pre-project loading: 78.7%
7. Facility: Cedar Creek Substation – Monroe Ne Substation 345.0 kV
 Contingency: Monroe Ne – Co Line Rd 500.0 kV
 Loading: 102.5% (Rating: 1,587 MVA)
 DFAX: 8.0%
 Pre-project loading: 101.7%
8. Facility: Gulfgate – Wake 138.0 kV
 Contingency: N Anna 345/138 kV Transformer
 Loading: 102.2% (Rating: 189 MVA)
 DFAX: 37.6%
 Pre-project loading: 71.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Acec Quincy – Ni Riv Dp 138.0 kV	Rebuild line with higher-capacity conductor	\$4,467,000	100% (\$4,467,000)
Acec Quincy Substation – Parsons Street 138.0 kV	Rebuild line with higher-capacity conductor	\$2,898,000	100% (\$2,898,000)
Parons St – Todds Tavern Dp 138.0 kV	Rebuild line with higher-capacity conductor	\$4,544,000	100% (\$4,544,000)
Chancellor – Ni Riv Dp 138.0 kV	Rebuild line with higher-capacity conductor	\$2,717,000	72% (\$1,969,000)
Gulfgate Substation – Story City Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,141,000	36% (\$1,513,000)
Chancellor Substation – Story City 138.0 kV	Reconductor line	\$829,000	41% (\$339,000)
Cedar Creek Substation – Monroe Ne Substation 345.0 kV	Reconductor line	\$12,293,000	1% (\$92,000)
Gulfgate Substation – Wake 138.0 kV	Reconductor line	\$2,707,000	36% (\$989,000)
	Total allocated to project		\$16,811,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.